

INDIRECT OPHTHALMOSCOPE LASER TREATMENT AS SUPPLEMENT TO ¹²⁵I PLAQUE THERAPY FOR CHOROIDAL MELANOMA*

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INTRODUCTION

EPISCLERAL PLAQUE RADIATION THERAPY HAS BEEN USED EXTENSIVELY in the management of primary malignant melanomas of the choroid and ciliary body during the past half century.¹ However, only a few choroidal melanomas regress completely to flat, atrophic scars following plaque irradiation.²⁻⁸ Most plaque-treated melanomas regress incompletely and persist for years as minimally to moderately shrunken residual lesions.²⁻⁸ Some of these residual lesions eventually reactivate and grow,^{5,8,9} and these tumors then require supplemental treatment.

Photocoagulation (xenon arc photocoagulation^{10,11} and laser therapy^{11,12}) has also been employed widely in the management of selected choroidal melanomas, although to a substantially lesser extent than has plaque therapy. Photocoagulation is generally believed to be an appropriate treatment for small, relatively thin tumors, especially those located near the optic disc in an extramacular location.¹⁰⁻¹² When photocoagulation is effective, the treated lesion is replaced by a flat, atrophic scar. Unfortunately, a substantial proportion of tumors treated by photocoagulation are not eradicated by this treatment and continue to grow.^{11,12}

Because the mechanisms of tumor destruction are different with plaque radiation therapy and photocoagulation, a number of experienced ophthalmologists have used, since at least the mid 1960s, both episcleral plaque radiation therapy and photocoagulation in a planned combination

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treatment to increase the likelihood of complete local tumor destruction in selected patients with a posterior uveal melanoma.¹³ Several clinicians have described good local results in a series of uncontrolled reports,¹³⁻¹⁶ but the true relative effectiveness of planned combined plaque therapy and photocoagulation versus plaque therapy alone is currently unknown.

We are indebted to King Lee, MD, of Kansas City, MO, who impressed us with his preliminary results of combined plaque and laser therapy¹⁶ in 1989 and encouraged us to embark on an investigation of his technique of combined therapy. We were not persuaded to do so, however, until we gained some clinical experience with an indirect ophthalmoscope argon laser¹⁷ and recognized that instrument's applicability to the treatment of choroidal melanomas. The indirect ophthalmoscope laser is an instrument that is easy to use by any ophthalmologist experienced in binocular indirect ophthalmoscopy. It can be used in both operating room and outpatient settings. It is applicable to the treatment of both peripheral and posterior choroidal tumors. Most important, the view during photocoagulation is satisfactory to permit treatment at low power on a continuous-exposure setting.

We have been using an argon laser indirect ophthalmoscope to supplement ¹²⁵I plaque radiotherapy in selected choroidal malignant melanomas since October 1990. The purposes of this paper are to summarize our preliminary results with planned combined plaque-laser therapy to date and to compare the local tumor regression results of this combined treatment with those of ¹²⁵I plaque therapy alone for similar tumors.

MATERIALS AND METHODS

PATIENTS

Between October 1990 and July 1991, we treated 26 eyes of 26 patients with planned sequential plaque radiation therapy and indirect ophthalmoscope argon laser therapy for a primary choroidal melanoma. All patients had undergone prospective assessment of multiple baseline clinical variables as part of an ongoing observational cohort study initiated in 1976. The methods of clinical examination, diagnosis, plaque therapy, and follow-up used in this group of patients have been published previously.^{18,19} Similarly, the variables assessed and the scales used in their mensuration have been published elsewhere.^{18,19}

One of the 26 patients was lost to follow-up shortly after initial laser treatment. For each of the remaining 25 patients in the combined plaque radiotherapy-laser photocoagulation group, We identified a patient from our concurrent and antecedent ¹²⁵I plaque radiotherapy group who was

was matched in terms of the following clinical variables: age at time of treatment (within ± 10 years), largest linear basal tumor diameter (within ± 3 mm), tumor thickness (within ± 1.5 mm), location of the posterior tumor margin relative to the optic disc (within ± 3 mm), and location of the posterior margin relative to the foveola (within ± 3 mm). Patient matching was performed in a masked fashion, using computer-generated listings of the five matching variables that did not include patient names or outcomes. The patient who most closely met all of the matching criteria was selected. Once a patient from the ^{125}I plaque group had been selected, that patient was removed from the list so as not to be selected again.

INVESTIGATIONAL APPROVAL

Prior to embarking on this investigation of combined plaque therapy and laser treatment, We obtained approval from the Institutional Review Board of Wills Eye Hospital to offer combined plaque and laser therapy to our patients with posterior uveal melanoma.

PLAQUE RADIOTHERAPY

All patients who agreed to undergo planned combined plaque therapy and photocoagulation first underwent conventional ^{125}I plaque therapy (apical target dose = 80 to 100 Gy). This treatment typically lasted 5 to 7 days. Patients having a tumor located 3 mm or less from the optic disc were treated with a posteriorly notched plaque. Laser treatment was started after completion of plaque therapy.

LASER THERAPY

All photocoagulation was performed with an indirect ophthalmoscope argon laser (Coherent, Inc) set on the green wavelength. The first laser treatment was usually performed immediately following plaque removal, but it was delayed for up to 3 to 4 months in patients with bullous retinal detachment to allow at least some of the subretinal fluid to be reabsorbed. The first laser treatment session entailed creation of a confluent, gray-white burn 2 to 3 burn diameters wide around the tumor base at the choroid-retinal pigment epithelial level (Fig 1A). Retinal blood vessels crossing over the barrier burn were usually not occluded during this therapy. Power settings for this therapy were typically in the range of 200 to 400 mW. The duration of exposure was "continuous" for every treatment.

The second and subsequent laser sessions (if needed) entailed retreatment of the tumor margin followed by confluent gray-white treatment to

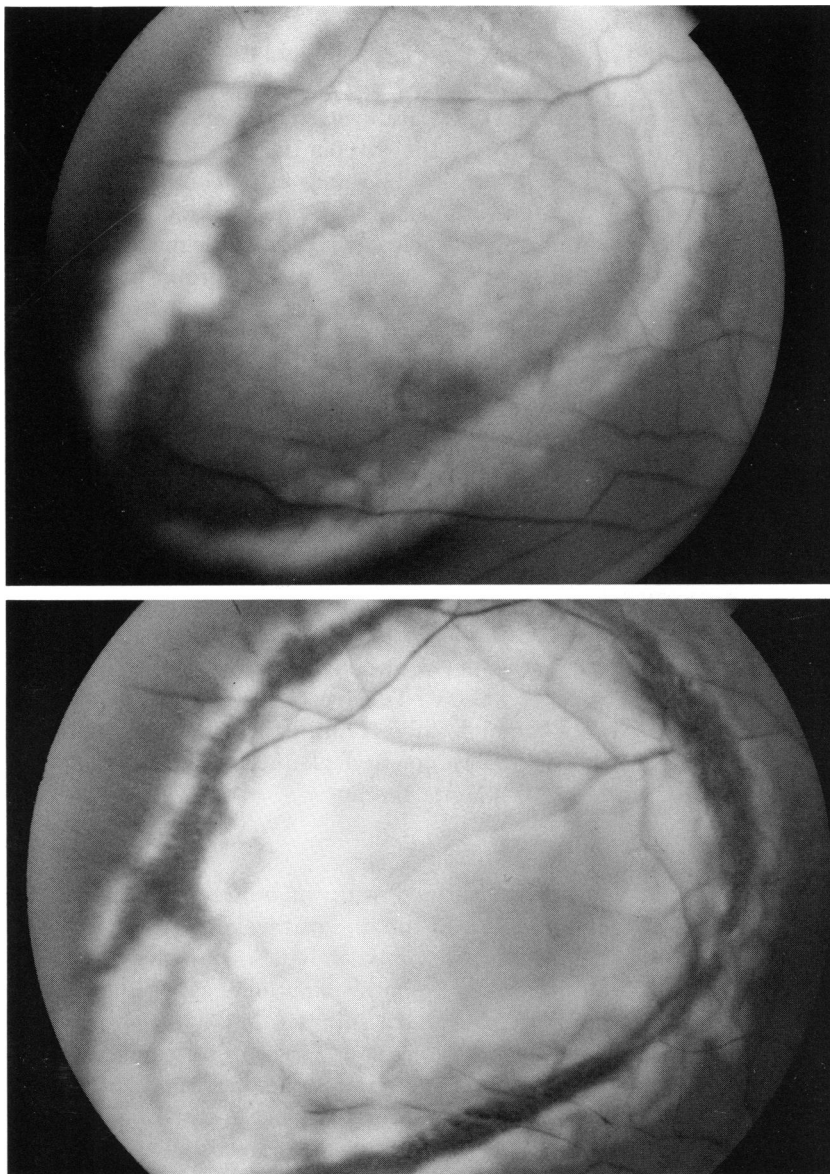


FIGURE 1

Choroidal malignant melanoma treated by radioactive ^{125}I plaque and indirect ophthalmoscope argon laser. A: Appearance of choroidal tumor immediately following initial session of laser therapy; confluent laser burn completely surrounds tumor base but does not extend onto surface of tumor. B: Appearance of same tumor immediately following second session of laser therapy; confluent laser burn covers entire surface of tumor.

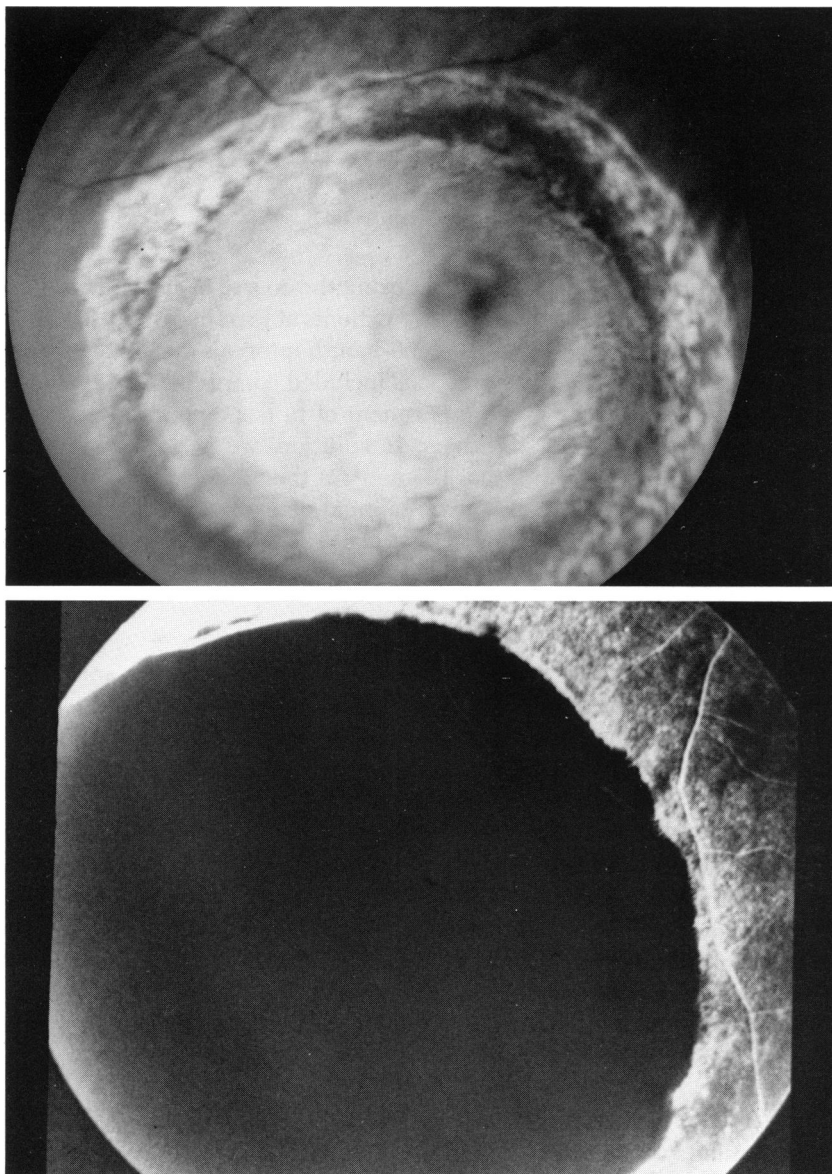


FIGURE 2

Desired end point of combined ^{125}I plaque therapy and indirect ophthalmoscopy argon laser for choroidal melanoma. A: Appearance of choroidal melanoma residue 6 months after plaque irradiation and two sessions of laser. B: Laminar venous phase frame of fluorescein angiogram showing well-defined nonfluorescence of treated tumor.

the entire surface of the tumor (Fig 1B). Laser power settings for the second and subsequent sessions were usually about the same as or slightly higher than for the initial session. Our end point for stopping laser therapy was a substantially flattened tumor with an atrophic circumbasal laser scar (Fig 2A) and complete nonfluorescence on fluorescein angiography (Fig 2B).

POSTTREATMENT FOLLOW-UP

Following plaque radiotherapy and during the course of subsequent laser photocoagulation, we reevaluated the patients at least every 1 to 3 months during the first 6 months and at 3- to 6-month intervals during the second 6 months. Each of these reevaluations included comprehensive binocular indirect ophthalmoscopy with assessment of tumor margins and an estimation of residual tumor thickness. In addition, we performed ophthalmic ultrasonography with tumor biometry at the visits closest to the 3, 6, and (when possible) 12-month posttreatment dates. We used the measurements of maximal tumor thickness at these follow-up visits to estimate local tumor regression.

COMPARISON OF STUDY GROUPS

We compared patients in the combined plaque-laser group and in the plaque group in terms of age, sex, eye, baseline tumor size, tumor location, and estimated radiation dose using chi-squared analysis (all variables).

ASSESSMENT OF RATES AND EXTENTS OF LOCAL TUMOR REGRESSION

We evaluated the relative local regression of the tumors in the two groups graphically by plotting the average tumor thickness in the groups over time following treatment (PATPLOT routine, MEDLOG Clinical Data Management System, Information Analysis Corp, Mountain View, CA). We also compared the mean thickness of matched tumors in the two groups at the 12-month interval using paired *t*-testing.

RESULTS

The patients in the two groups were remarkably similar in terms of baseline demographic features and tumor characteristics (Table I). Following plaque therapy, the patients in the combined plaque-laser group underwent from one to five laser treatments (median, 2 treatments; mean, 2.5 treatments) over a median postirradiation interval of 16.5 weeks. Four patients had one laser session, 10 patients had two, 9 patients

had three, 1 patient had four, and 1 patient had five.

Two of the 25 patients in the plaque radiotherapy group had documented local tumor relapse during the first postirradiation year. In contrast, none of the 25 patients in the combined plaque-laser group had local tumor relapse. Only one patient died during the first postirradiation year, and that patient (a patient in the plaque therapy group) died of a nonmelanoma cancer.

The local tumor regression appeared to be substantially more rapid and more complete in the patients treated by combined plaque radiation therapy and laser (Fig 3). The difference in mean thickness of the treated tumors in the two groups at 12 months posttreatment was statistically significant ($P = 0.003$, paired t -test).

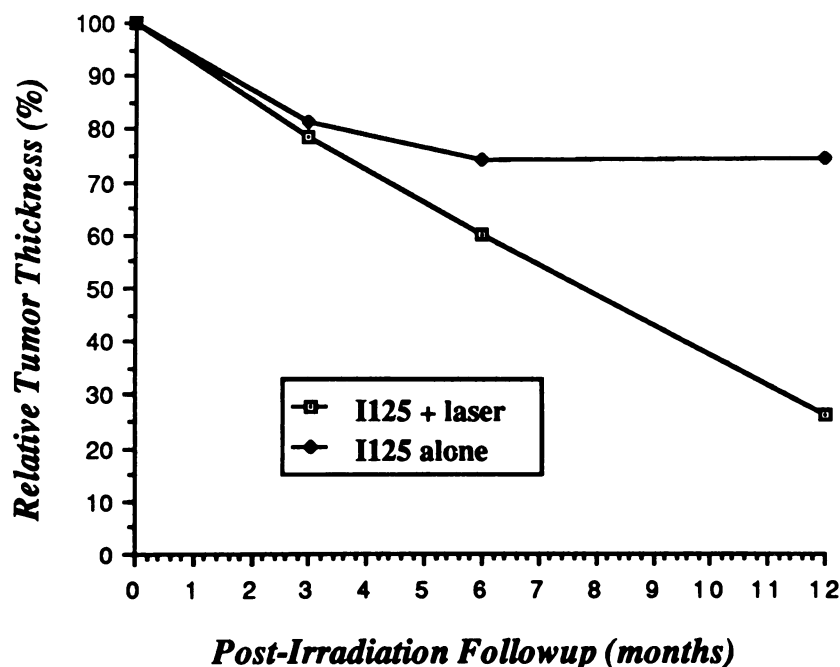


FIGURE 3

Graphic presentation of course of local tumor regression (mean percentage of initial tumor thickness) over time following ^{125}I plaque plus laser versus ^{125}I plaque alone.

TABLE I: BASELINE DEMOGRAPHIC FEATURES AND TUMOR CHARACTERISTICS OF MATCHED PATIENT GROUPS

VARIABLE AND CATEGORIES*	1231 PLAQUE PLUS LASER		1231 PLAQUE ALONE		CHI-SQUARE	P
	NO.	%	MEAN $\pm$	MEAN $\pm$		
Age						
≤ 57 yr	11	44		12	48	
> 57 yr	14	56		13	52	< 0.01
			56.2 $\pm$ 14.9		57.5 $\pm$ 12.8	> 0.5
Sex						
Male	10	40		13	52	
Female	15	60		12	48	0.32
						> 0.5
Eye						
Right	15	60		11	44	
Left	10	40		14	56	0.72
						0.40
Largest linear basal tumor diameter						
≤ 10 mm	10	40		9	36	< 0.01
			9.8 $\pm$ 3.1		9.8 $\pm$ 2.5	> 0.5
Tumor thickness						
≤ 4 mm	14	56		13	52	
> 4 mm	11	44		12	48	< 0.01
			4.0 $\pm$ 1.7		3.9 $\pm$ 1.3	> 0.5
Location of anterior tumor margin (relative to ocular equator)						
At or posterior to equator	18	72		14	56	
Anterior to equator	7	28		11	44	0.78
						0.38

Location of posterior tumor margin (relative to optic disc)					
Within 3 mm from disc	8	32	11	44	
Over 3 mm from disc	17	68	14	56	
Location of posterior tumor margin (relative to foveola)					0.34
Within 3 mm from foveola	13	52	13	52	
Over 3 mm from foveola	12	48	12	48	
Radiation dose to tumor apex					0.00
≤ 95 Gy	12	48	13	52	
> 95 Gy	13	52	12	48	
			101.1 ± 16.3	93.1 ± 10.4	< 0.01
Radiation dose to tumor base					> 0.5
≤ 340 Gy	15	60	11	44	
> 340 Gy	10	40	14	56	
			336.6 ± 50.2	338.6 ± 60.2	0.72
					0.40

*Categories of continuous numeric variables are dichotomized at the median values.

†Mean values are shown only for continuous numeric variables.

DISCUSSION

When we initially presented our results of this combined therapy at the 1991 Retina Society meeting, we expressed the opinion that combined plaque-laser treatment appeared to induce more profound and faster local tumor regression than did plaque therapy alone. We devised this study to compare the rate and extent of local tumor regression following planned combined plaque therapy and indirect ophthalmoscope laser treatment with that following ^{125}I plaque radiotherapy alone. The results of this comparative analysis appear to confirm our clinical impression that ^{125}I plaque therapy combined with indirect ophthalmoscope laser treatment causes faster and more complete local regression of choroidal melanomas than does ^{125}I plaque therapy alone.

The concept of combining photocoagulation with episcleral plaque radiotherapy for the treatment of choroidal malignant melanomas is not new. Photocoagulation has been used for many years as a secondary treatment in eyes with limited local tumor relapse or incomplete local tumor regression following plaque therapy. One experienced group, for example, has treated approximately 20% to 30% of its ^{106}Ru plaque-treated eyes with supplemental xenon arc photocoagulation or laser therapy for residual or recurrent local choroidal melanoma.¹⁰ This secondary application of photocoagulation for recurrent or nonregressing choroidal melanomas, however, is not the subject of this paper.

Photocoagulation has also been used as a planned supplement to episcleral plaque therapy of choroidal melanomas for a number of years. Boniuk and Cohen¹³ first described planned photocoagulation (xenon arc) as a supplement to gold plaque radiotherapy (40 eyes) in 1978. Subsequently, Zygulska-Mach and associates¹⁴ (1983) reported their use of xenon arc photocoagulation as a planned supplement to ^{60}Co plaque radiotherapy (77 eyes), and Moura and associates¹⁵ (1985) described their experience with argon laser photocoagulation as a supplement to gold plaque radiotherapy (21 eyes). More recently, Lee and co-workers¹⁶ reported their experience with dye laser therapy as a supplement to ^{125}I plaque treatment (17 eyes). All of these investigators were pleased with the results of their combined-modalities therapy and have recommended more widespread use of this form of therapy.

In spite of the favorable local tumor response in patients treated by combined ^{125}I plaque radiotherapy and indirect ophthalmoscope laser therapy, a number of important questions remain about this method of treatment. Will the combined treatment really reduce the rate of local tumor relapse? We expect so, but we really do not know. Will the combined treatment improve the survival prognosis? We hope so, but we

have no good reason at this time to believe that our combined-modalities therapy will substantially alter the expected mortality pattern of our patients. Will the observed improvements (if any) be worth the cost? This may be the most important question of all. If the treatment does not improve survival prognosis substantially, is the faster and more complete local regression of choroidal melanomas observed with this therapy worth the expected reduction in local tumor relapse of a relatively small fraction of patients? We simply do not know. Finally, are there some subgroups of patients with posterior uveal melanoma who would benefit more than others from this combined-methods therapy? There may well be, but there are simply too few patients yet treated to allow one to even consider this question.

Recognizing the serious limitations of my current experience in terms of number of patients treated and their length of follow-up, we make no strong recommendation for or against planned combined ^{125}I plaque radiotherapy and indirect ophthalmoscope laser therapy. However, on the basis of our experience and that of other cited clinicians, we believe that this combined-therapies approach has sufficient potential merit to warrant additional evaluation by other groups, a thorough public discussion, and consideration for future use in patients with choroidal melanoma.

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DISCUSSION

DR ROBERT M. ELLSWORTH. Doctor Augsburger properly stresses two points. First, this type of combined therapy is not new. Doctor Boniuk, was an early champion of use of combined radiation plaques and photocoagulation, and in 1978 reported on 40 patients treated with radioactive gold seeds encased in platinum and placed on a silicone rubber plaque. These were individually designed to conform to tumor size, and the gamma ray dose varied from 7,000 to 24,000 rads at the apex. While generally these were smaller tumors, Doctor Boniuk listed his criterion as a diameter of less than 10 mm, although tumors up to 15 mm were treated with a height of less than 12 diopters.

In 1980, Zygulska-Mach and associates reported on 174 patients with uveal melanomas treated in Poland using combined ^{60}Co plaques and xenon-arc photocoagulation. This is a technique that we have used intermittently over the past 30 years with considerable success.

Doctor Augsburger's second point is that combined therapy makes eminently good sense. The mechanism of action is different, and each therapy has attractive aspects. Brachytherapy has been criticized because of uneven dose distribution, and proton-beam radiation has been favored because the depth-dose characteristics are more uniform. Perhaps, however, the high base dose is just what is needed to control these tumors. There generally has not been any scleral problem, but occasional necrosis is seen when the base dose is more than 50,000 cG. Theoretically, the tumor cells at the apex of the tumor may be relatively anoxic

compared with the rest of the tumor, and therefore radioresistant. These less radiated, and possibly anoxic, cells are in an ideal position for light coagulation at the completion of radiation therapy.

The failure of radiation therapy to locally control tumor in any situation may well be related to a small percentage of cells, perhaps less than 1%, that have unusual radioresistance for reasons that are not completely understood. This small fraction of cells could remain viable following plaque therapy and should be easily destroyed using the heat of light coagulation. In our hands, we have used the xenon arc, which became available in 1958 and had sufficient energy. It must be stressed that these light coagulation treatments are quite intense and destructive and must be repeated until the scar is flat.

We experimented with the early indirect ophthalmoscopes adapted for argon laser coagulation and originally felt that there was not enough energy for this purpose. The newer indirect ophthalmoscope argon laser made by Coherent, Inc, apparently has plenty of energy for this purpose, and it is certainly a much easier instrument to manipulate.

We have, in essence, two modalities that are relatively satisfactory alone. We have treated mushroom-shaped malignant melanomas to a height of 8 mm or so with xenon-arc coagulation alone, although, again, it must be emphasized that repeated treatments at intervals of 3 to 4 weeks and very heavy coagulation are necessary. During the sessions, steam bubbles may build up in the tumor and explode, scattering pigmented cells into the vitreous. Hemorrhage can follow immediately, but if, under direct observation, this does begin to occur, it can be coagulated by further light coagulation. I have never seen seeding elsewhere in the retina from these cells, which are putatively destroyed in the explosion.

Melanomas, either elevated or flat, around the nerve head, may present special dangers. Doctor Reese stressed the insidious nature of flat melanomas around the nerve head that could migrate down through the border tissue along the nerve. Such lesions would be difficult to control by either radiation or light coagulation, and perhaps the improved computed tomographic and magnetic resonance imaging thin-section techniques may enable us to detect these patients, who would be far better off with immediate enucleation.

Doctor Augsburger valiantly attempted to control the series with earlier cases treated by ^{125}I brachytherapy alone, and this is not, and cannot be, perfect. Moreover, the follow-up period is too short to draw conclusions as far as mortality is concerned. Nonetheless, this combined therapy makes sense and should be considered as a standard approach in the future.

DR W. BANKS ANDERSON. It is my understanding that radiation sensitivity of tumors is a function of tumor oxygenation and similarly it is important to stress that the laser treatment should be done after the radiation not before.

DR JAMES J. AUGSBURGER. Thank you, Doctor Ellsworth and Doctor Anderson. Let me first respond to Doctor Anderson's comment. I agree wholeheartedly. If indirect ophthalmoscope laser treatment is used to supplement ^{125}I plaque irra-

diation in a planned combination therapy, the plaque treatment should always be performed first for the reason Doctor Anderson mentioned. Regarding Doctor Ellsworth's comments, let me emphasize that the technique of indirect ophthalmoscope laser therapy I described and illustrated in my presentation is substantially different from the usual method of photocoagulation Doctor Ellsworth mentioned. In "conventional" laser treatment of choroidal melanomas, the surgeon uses high power, short duration laser exposures to create multiple overlapping, intensely white burns. In my treatment, the surgeon uses relatively low power, long duration (continuous setting) laser exposures to produce a confluent, slowly developing gray-white burn. When conventional high power-short duration laser treatment is employed on the surface of the tumor during the second and any subsequent sessions, it commonly causes (1) focal ruptures of Bruch's membrane accompanied by an audible "pop" and liberation of pigment into the overlying vitreous; (2) acute rupture of retinal or choroidal blood vessels resulting in bleeding from the tumor surface into the overlying vitreous; and (3) abrupt wrinkling of the retina. In contrast, with my low power-long exposure indirect ophthalmoscope laser technique, I have not yet encountered any of these problems intraoperatively. Based on my experience to date, I believe that low power-long exposure laser treatment of choroidal melanomas using an indirect ophthalmoscope laser may prove to be the photocoagulation method of choice for these tumors.